Exact Spectral Gap and Quantitative Convergence Rates for the Kuramoto Self-Consistency Equation

Abstract

We study the self-consistency map $\Phi(r) = \int \alpha^*(\gamma, K, r)\,\mathrm{d}\mu(\gamma)$ that determines the equilibrium order parameter of the mean-field Kuramoto model on the Ott–Antonsen manifold. Here μ is a probability measure on a positive dissipation parameter $\gamma > 0$ satisfying $\int (1/\gamma)\,\mathrm{d}\mu \in (0, \infty)$. We prove:

(i) an exact closed-form expression for the derivative $\Phi'(r^*)$ and the spectral gap $1 - \Phi'(r^*) = \int K^3 r^{*2}/(D_\gamma(\gamma + D_\gamma)^2)\,\mathrm{d}\mu$ where $D_\gamma = \sqrt{\gamma^2 + K^2 r^{*2}}$;

(ii) two-sided critical slowing down: the spectral gap satisfies $1 - \lambda = \Theta(K - K_c)$ at the onset of synchronization (under bounded γ);

(iii) geometric convergence of Picard iterates to r^* from both above and below, with explicitly computable contraction rates;

(iv) strictly subcritical exponential decay: $\Phi^n(r_0) \le (K/K_c)^n r_0 \to 0$ for $K < K_c$.

The results are accompanied by a Lean 4 formalization (Mathlib v4.30.0-rc1, 0 `sorry`, no project-specific axioms); the repository link, commit hash, and exact Lean declaration names are given in Section 12. The exact spectral gap formula and the identification of the contraction rate with $\Phi'(r^*)$ appear to be new.

Contents

1 Introduction

1.1 Background

The Kuramoto model [6, 7] of coupled oscillators exhibits a phase transition from incoherence to partial synchronization at a critical coupling strength K_c. On the Ott–Antonsen manifold [10], the infinite-dimensional PDE reduces to a family of ODEs parametrized by frequency, and the equilibrium order parameter r^* satisfies the self-consistency equation

$$r^* = \Phi(r^*) := \int \alpha^*(\gamma, K, r^*) \, \mathrm{d}\mu(\gamma), \tag{1}$$

where $\alpha^*(\gamma, K, r)$ is the equilibrium locking amplitude of an oscillator with dissipation parameter γ at coupling K and mean-field r. The explicit formula is

$$\alpha^*(\gamma, K, r) = \frac{-\gamma + \sqrt{\gamma^2 + K^2 r^2}}{Kr}. \tag{2}$$

The existence of a critical coupling K_c and the qualitative bifurcation structure have been understood since Kuramoto's original work [6]. Strogatz and Mirollo [12] established spectral stability of incoherence for the Lorentzian case and analyzed the self-consistency equation in detail. Crawford [2, 3] developed center manifold analysis for the onset instability, identifying the critical slowing down phenomenon qualitatively via amplitude expansions. Chiba [1] proved nonlinear stability of the partially locked state using generalized spectral theory on rigged Hilbert spaces. Dietert [4] established sharp nonlinear Landau damping for both subcritical and supercritical regimes. Strogatz [11] provides a comprehensive survey of the Kuramoto model literature through 2000.

Despite this extensive literature, the *quantitative* convergence theory of the self-consistency iteration—the explicit rate at which the Picard scheme $r_{n+1} = \Phi(r_n)$ converges, and the exact dependence of this rate on system parameters—has not been developed for general distributions. The self-consistency equation is routinely solved numerically by iteration, yet rigorous a priori convergence guarantees with computable error bounds have been absent.

1.2 Main contribution

Our central result is an exact formula for the derivative of the self-consistency map at its fixed point:

$$\lambda := \Phi'(r^*) = \int \frac{K\gamma}{D_\gamma(\gamma + D_\gamma)}\,\mathrm{d}\mu, \qquad D_\gamma = \sqrt{\gamma^2 + K^2 r^{*2}}, \tag{3}$$

with $0 < \lambda < 1$ strictly whenever $r^* > 0$ is a fixed point. The spectral gap $1 - \lambda$ admits the exact integral representation

$$1 - \lambda = \int \frac{K^3 r^{*2}}{D_\gamma(\gamma + D_\gamma)^2}\,\mathrm{d}\mu > 0. \tag{4}$$

This identity arises from the pointwise algebraic relation

$$\frac{K}{\gamma + D} - \frac{K\gamma}{D(\gamma + D)} = \frac{K^3 r^2}{D(\gamma + D)^2},$$

combined with the slope identity $\int K/(\gamma + D)\,\mathrm{d}\mu = 1$ at the fixed point (Section 3).

From (4), we derive:

- **Critical slowing down** (under bounded γ): $1 - \lambda = O(K - K_c)$ as $K \to K_c^+$.
- **Geometric convergence from above**: For $r_0 > r^*$, the Picard iterates satisfy $0 \le r_n - r^* \le \lambda^n (r_0 - r^*)$.
- **Subcritical global decay**: For $K < K_c$, the bound $\Phi(r) \le (K/K_c)\, r$ gives $\Phi^n(r_0) \le (K/K_c)^n r_0 \to 0$.

1.3 What is and is not proved

We distinguish results by the assumptions they require:

1. **Minimal assumptions (H1)–(H3) only**: the slope identity (Lemma 3.2), the spectral gap identity (4), the contraction $\lambda < 1$, the sign lemma (Lemma 3.3), existence and uniqueness of r^* for $K > K_c$, the linear bound $\Phi(r) \le (K/K_c)r$, subcritical decay, Picard convergence (from above and below).
2. **Bounded γ (H4)**: two-sided bounds on $1 - \lambda$, the critical slowing down estimate.
3. **Finite mean (H5):** $\int \gamma\,\mathrm{d}\mu < \infty$: the contraction rate upper bound $\lambda \le \mathbb{E}[\gamma]/(K r^{*2})$.
4. **Third inverse moment**: $\int 1/\gamma^3\,\mathrm{d}\mu < \infty$ for the lower bound on r^* in the square-root law.

The geometric convergence rate λ^n is proved only for initial data $r_0 > r^*$ (Theorem 6.2). For $r_0 < r^*$, we prove monotone convergence $r_n \nearrow r^*$ (Theorem 6.3) but do not establish a geometric rate.

1.4 Relation to prior work

The qualitative bifurcation diagram (existence and uniqueness of r^* for $K > K_c$) is classical for the Lorentzian distribution [12] and has been extended to general distributions via the intermediate value theorem applied to the slope function $S(r) = \Phi(r)/r$, which is strictly decreasing (see Lemma 3.3 below). The Ott–Antonsen reduction [10] provides the explicit equilibrium formula (2). Crawford [2, 3] used amplitude expansions to identify the $\beta = 1/2$ critical exponent and the qualitative slowing down at onset.

Our contribution is complementary: we study the self-consistency map Φ as a nonlinear operator on $(0, \infty)$ and establish its quantitative contraction properties. The exact formula (4) appears not to have been written down previously, though the equilibrium profile and its derivative are individually known in the literature. Crawford's amplitude expansion [2] gives the critical exponent but not the quantitative convergence rate of the iteration; Chiba's generalized spectral analysis [1] concerns the PDE eigenvalues, not the map derivative. The relationship between $\Phi'(r^*)$ and the PDE stability exponent is discussed in Section 13.

We emphasize that linearization of the OA system around the partially locked state gives eigenvalues related to, but distinct from, $1 - \lambda$: the self-consistency map derivative governs the *iterative* convergence of the fixed-point equation, while the PDE eigenvalue governs the *dynamical* stability.

1.5 Formal verification

The results in this paper have been formally verified in Lean 4 [8] using Mathlib [9] v4.30.0-rc1. The formalization is available at:

`https://github.com/taejun-song/kuramoto-lean`, commit `e86b442`

Lean toolchain: `leanprover/lean4:v4.30.0-rc1`. Mathlib: v4.30.0-rc1. The exact correspondence between paper theorems and Lean declarations is given in Section 12. The main file `TightLipschitz.lean` contains approximately 580 lines of proof.

2 Setup and notation

2.1 The model

Let $(\Omega, \mathcal{F}, \mu)$ be a probability space and $\gamma : \Omega \to (0, \infty)$ a measurable function. We call γ the *dissipation parameter*.

Remark 2.1 (On the parameter γ)**.** *The parameter γ is* not *the natural frequency ω of the standard Kuramoto model (which takes values in $\mathbb{R}$ and can be negative). In the Ott–Antonsen reduction with symmetric frequency density $g(\omega)$ and real-valued initial data, the complex OA equation reduces to the scalar equation*

$$\dot{\alpha}(\omega, t) = -\gamma(\omega)\,\alpha + \tfrac{K}{2}\,r(t)(1 - \alpha^2), \tag{5}$$

where $\gamma(\omega) > 0$ encodes the effective dissipation/detuning. For the Cauchy–Lorentz distribution $g(\omega) = \gamma_0/(\pi(\omega^2 + \gamma_0^2))$, residue evaluation gives $\gamma = \gamma_0$ (constant). The probability space (Ω, μ) parametrizes the oscillator population: $\gamma(\omega)$ assigns to each oscillator its positive dissipation rate, and μ is the distribution of these rates. Our theorems take $\gamma > 0$ and the self-consistency equation (1) *as the starting point; the derivation of γ from the underlying frequency density is a separate reduction step.*

2.2 Standing assumptions

(H1) $\gamma(\omega) > 0$ for all $\omega \in \Omega$ (positive dissipation);
(H2) $\int (1/\gamma)\,\mathrm{d}\mu < \infty$ (finite harmonic mean);
(H3) $\int (1/\gamma)\,\mathrm{d}\mu > 0$ (non-degeneracy).
Conditions (H2)–(H3) ensure that K_c is finite and positive.

Remark 2.2 (On the integrability hypothesis). *The condition $\int(1/\gamma)\,\mathrm{d}\mu < \infty$ is an assumption on the distribution of γ near zero,* not *a moment condition on γ itself. It is strictly weaker than γ being bounded away from zero, and unrelated to the condition $\int \gamma\,\mathrm{d}\mu < \infty$ (finite first moment). For example, $\mu =$ Uniform$(0,1)$ on positive γ fails (H2) since $\int_0^1 \gamma^{-1}\,\mathrm{d}\gamma = \infty$, while $\mu =$ Uniform$(1,2)$ satisfies all hypotheses trivially.*

The framework covers any probability distribution on positive dissipation parameters satisfying (H2): point masses (Lorentzian residue case), finite mixtures, bounded positive distributions, and positive unbounded distributions with sufficiently little mass near zero.

2.3 Definitions

Definition 2.3 (Critical coupling).

$$K_c := \frac{2}{\int(1/\gamma)\,\mathrm{d}\mu}. \tag{6}$$

Definition 2.4 (Equilibrium profile). For $\gamma, K, r > 0$:

$$\alpha^*(\gamma, K, r) := \frac{-\gamma + \sqrt{\gamma^2 + K^2 r^2}}{Kr} = \frac{Kr}{\gamma + \sqrt{\gamma^2 + K^2 r^2}}. \tag{7}$$

The second form (rationalized) is used throughout.

Definition 2.5 (Self-consistency map).

$$\Phi(r) := \int \alpha^*(\gamma(\omega), K, r)\,\mathrm{d}\mu(\omega). \tag{8}$$

We write $D = D_\gamma(r) := \sqrt{\gamma^2 + K^2 r^2}$.

Proposition 2.6 (Basic properties). *Under (H1), for all $K, r > 0$:*
(a) $0 < \alpha^*(\gamma, K, r) < 1$*;*
(b) α^* *is strictly increasing in r and strictly decreasing in γ;*
(c) $\alpha^*(\gamma, K, r) < Kr/\gamma$*;*
(d) $\alpha^*(\gamma, K, r) = Kr/(\gamma + D) \le Kr/(2\gamma)$ *(since $D \ge \gamma$).*

3 The slope identity and sign lemma

3.1 Slope factorization

The self-consistency map factors as $\Phi(r) = S(r)\cdot r$ where

$$S(r) = \int \frac{K}{\gamma + D_\gamma(r)}\,\mathrm{d}\mu \tag{9}$$

is the *slope function.*

Lemma 3.1 (Properties of the slope function). *Under (H1)–(H3), $S : (0,\infty) \to (0,\infty)$ is continuous, strictly decreasing, and satisfies*

$$\lim_{r\downarrow 0} S(r) = \frac{K}{K_c}, \qquad \lim_{r\to\infty} S(r) = 0.$$

Proof. Continuity. For each ω, $r \mapsto K/(\gamma(\omega) + D_\gamma(r))$ is continuous. The pointwise bound $K/(\gamma+D) \le K/(2\gamma)$ and $\int K/(2\gamma)\,\mathrm{d}\mu = K/K_c < \infty$ (by (H2)) provide an integrable dominator. Dominated convergence gives continuity of S.

Strict decrease. For $r_1 < r_2$, $D_\gamma(r_1) < D_\gamma(r_2)$ for every ω, so $K/(\gamma + D_\gamma(r_1)) > K/(\gamma + D_\gamma(r_2))$ pointwise. Since μ is a nonzero probability measure and the strict inequality holds on a set of full measure, integration preserves the strict inequality: $S(r_1) > S(r_2)$.

Limits. As $r \downarrow 0$: $D_\gamma(r) \to \gamma$, so $K/(\gamma + D) \to K/(2\gamma)$. Dominated convergence gives $S(0^+) = \int K/(2\gamma)\,\mathrm{d}\mu = K/K_c$. As $r \to \infty$: $D_\gamma(r) \to \infty$, so each integrand $\to 0$; dominated convergence gives $S(r) \to 0$. ☐

Lemma 3.2 (Slope identity at the fixed point)**.** *If $r^* > 0$ satisfies $\Phi(r^*) = r^*$, then $S(r^*) = 1$.*

Proof. From $\alpha^* = Kr/(\gamma + D)$: $\Phi(r^*) = r^* S(r^*)$. Since $\Phi(r^*) = r^*$ and $r^* > 0$, divide to get $S(r^*) = 1$. ☐

3.2 Sign of $\Phi(r) - r$

The strict monotonicity of S determines the sign of $\Phi(r) - r$ on either side of the fixed point.

Lemma 3.3 (Sign lemma)**.** *Let $r^* > 0$ with $\Phi(r^*) = r^*$. Then for all $r > 0$:*

$$r < r^* \implies \Phi(r) > r \qquad \textit{(the map overshoots below } r^*\textit{)}, \tag{10}$$

$$r > r^* \implies \Phi(r) < r \qquad \textit{(the map undershoots above } r^*\textit{)}. \tag{11}$$

Proof. Since S is strictly decreasing and $S(r^*) = 1$: for $r < r^*$, $S(r) > S(r^*) = 1$, so $\Phi(r) = rS(r) > r$. For $r > r^*$, $S(r) < 1$, so $\Phi(r) = rS(r) < r$. ☐

Remark 3.4. *The sign lemma, combined with Lemma 3.1, implies the qualitative bifurcation structure. For $K > K_c$: $S(0^+) = K/K_c > 1$ and $S(r) \to 0$, so by the intermediate value theorem there is a unique $r^* > 0$ with $S(r^*) = 1$. For $K \le K_c$: $S(r) < S(0^+) = K/K_c \le 1$ for all $r > 0$, so no positive fixed point exists. See Theorem 3.5.*

3.3 Existence and uniqueness

Theorem 3.5 (Bifurcation)**.** *Under (H1)–(H3):*

(a) If $K \le K_c$, then $\Phi(r) < r$ for all $r > 0$. No positive fixed point exists.

(b) If $K > K_c$, there exists a unique $r^* \in (0, 1)$ with $\Phi(r^*) = r^*$.

Proof. Part (a): For $K \le K_c$, $S(r) < S(0^+) = K/K_c \le 1$ for all $r > 0$, so $\Phi(r) = rS(r) < r$. (At $K = K_c$ the inequality $S(r) < 1$ is strict for $r > 0$ since $D_\gamma(r) > \gamma$ implies $K/(\gamma+D) < K/(2\gamma)$.)

Part (b): By Lemma 3.1, S is continuous and strictly decreasing on $(0, \infty)$ with $S(0^+) = K/K_c > 1$ and $S(r) \to 0$. By the intermediate value theorem, there exists a unique r^* with $S(r^*) = 1$, equivalently $\Phi(r^*) = r^*$. The bound $r^* < 1$ follows from $\alpha^* < 1$. ☐

3.4 Exact derivative of the equilibrium profile

Theorem 3.6 (Exact derivative)**.** *For $\gamma, K, r > 0$:*

$$\frac{\partial}{\partial r}\alpha^*(\gamma, K, r) = \frac{K\gamma}{D(\gamma + D)}, \qquad D = \sqrt{\gamma^2 + K^2 r^2}. \tag{12}$$

This derivative is strictly decreasing in r (since D and $\gamma + D$ are increasing in r).

Proof. Differentiating $\alpha^* = Kr/(\gamma + D)$ with $\partial D/\partial r = K^2 r/D$:

$$\frac{\partial \alpha^*}{\partial r} = \frac{K(\gamma + D) - Kr \cdot K^2 r/D}{(\gamma + D)^2} = \frac{K[\gamma D + D^2 - K^2 r^2]}{D(\gamma + D)^2} = \frac{K[\gamma D + \gamma^2]}{D(\gamma + D)^2} = \frac{K\gamma}{D(\gamma + D)},$$

using $D^2 - K^2 r^2 = \gamma^2$. $\square$

Theorem 3.7 (Tight Lipschitz bound). *For $0 < r_1 < r_2$:*

$$\alpha^*(\gamma, K, r_2) - \alpha^*(\gamma, K, r_1) \leq \frac{K\gamma}{D_1(\gamma + D_1)} \cdot (r_2 - r_1),$$

where $D_1 = \sqrt{\gamma^2 + K^2 r_1^2}$.

Proof. By the mean value theorem, $\alpha^*(r_2) - \alpha^*(r_1) = (\partial_r \alpha^*)(c) \cdot (r_2 - r_1)$ for some $c \in (r_1, r_2)$. Since $\partial_r \alpha^*$ is decreasing in r (Theorem 3.6), $(\partial_r \alpha^*)(c) \leq (\partial_r \alpha^*)(r_1) = K\gamma/(D_1(\gamma + D_1))$. $\square$

Theorem 3.8 (Derivative strictly less than slope). *For all $\gamma, K, r > 0$:*

$$\frac{K\gamma}{D(\gamma + D)} < \frac{K}{\gamma + D}.$$

Proof. Divide both sides by $K/(\gamma + D) > 0$ to get $\gamma/D < 1$, which holds since $D > \gamma$ for $r > 0$. $\square$

4 Spectral gap: exact formula and bounds

4.1 The contraction rate

Lemma 4.1 (Differentiation under the integral). *Under (H1)–(H3), Φ is differentiable on $(0, \infty)$ with*

$$\Phi'(r) = \int \frac{\partial}{\partial r} \alpha^*(\gamma, K, r)\, \mathrm{d}\mu = \int \frac{K\gamma}{D(\gamma + D)}\, \mathrm{d}\mu.$$

Proof. For each ω, $r \mapsto \alpha^*(\gamma(\omega), K, r)$ is differentiable with derivative $K\gamma/(D(\gamma + D))$ (Theorem 3.6). For all r in a neighborhood of any $r_0 > 0$, this derivative satisfies $K\gamma/(D(\gamma + D)) \leq K/(\gamma + D) \leq K/(2\gamma)$, and $K/(2\gamma)$ is μ-integrable by (H2). The Leibniz integral rule (via dominated convergence) gives $\Phi'(r) = \int \partial_r \alpha^*\, \mathrm{d}\mu$. $\square$

Theorem 4.2 ($\lambda < 1$ at any positive fixed point). *If $r^* > 0$ satisfies $\Phi(r^*) = r^*$, then*

$$\lambda := \Phi'(r^*) = \int \frac{K\gamma(\omega)}{D_\gamma(\gamma(\omega) + D_\gamma)}\, \mathrm{d}\mu < 1.$$

Proof. The equality $\lambda = \Phi'(r^*)$ holds by Lemma 4.1. By Theorem 3.8, pointwise $K\gamma/(D(\gamma + D)) < K/(\gamma + D)$. Both integrands are positive and integrable (the right-hand side by Lemma 3.2). Hence $\lambda < \int K/(\gamma + D)\, \mathrm{d}\mu = 1$. $\square$

4.2 Exact spectral gap

Theorem 4.3 (Spectral gap identity). *At a fixed point $r^* > 0$:*

$$1 - \lambda = \int \frac{K^3 r^{*2}}{D_\gamma(\gamma + D_\gamma)^2}\, \mathrm{d}\mu. \tag{13}$$

Proof. The pointwise algebraic identity (using $D^2 = \gamma^2 + K^2 r^2$):

$$\frac{K}{\gamma + D} - \frac{K\gamma}{D(\gamma + D)} = \frac{K(D - \gamma)}{D(\gamma + D)} = \frac{K(D^2 - \gamma^2)}{D(\gamma + D)(D + \gamma)} = \frac{K^3 r^2}{D(\gamma + D)^2}.$$

Integrating and applying Lemma 3.2:

$$1 - \lambda = \int \frac{K}{\gamma + D}\, \mathrm{d}\mu - \int \frac{K\gamma}{D(\gamma + D)}\, \mathrm{d}\mu = \int \frac{K^3 r^{*2}}{D(\gamma + D)^2}\, \mathrm{d}\mu. \qquad \square$$

4.3 Two-sided bounds (bounded γ)

The following bounds require the additional assumption:

$$0 < \gamma_{\min} \le \gamma(\omega) \le \gamma_{\max} \quad \text{for all } \omega. \tag{H4}$$

Theorem 4.4 (Two-sided bounds on the spectral gap)**.** *Under* (H4)*, with* $D_{\min} = \sqrt{\gamma_{\min}^2 + K^2 r^{*2}}$ *and* $D_{\max} = \sqrt{\gamma_{\max}^2 + K^2 r^{*2}}$*:*

$$\frac{K^3 r^{*2}}{D_{\max}(\gamma_{\max} + D_{\max})^2} \ \le\ 1 - \lambda \ \le\ \frac{K^3 r^{*2}}{D_{\min}(\gamma_{\min} + D_{\min})^2}. \tag{14}$$

Proof. The integrand $K^3 r^{*2}/(D(\gamma+D)^2)$ is decreasing in γ (since D and $\gamma+D$ increase with γ). Evaluating at $\gamma_{\max}$ gives the lower bound and at $\gamma_{\min}$ the upper bound. Since μ is a probability measure, integration preserves the pointwise bounds. □

5 Critical slowing down

Near $K = K_c$, the equilibrium r^* vanishes as $r^* = \Theta(\sqrt{K - K_c})$ (critical exponent $\beta = 1/2$). Since the spectral gap (13) is $O(r^{*2})$, it vanishes as $O(K - K_c)$.

Theorem 5.1 (Square-root law)**.** *Under* (H4)*, for* $K > K_c$ *with fixed point* $r^* \in (0, 1)$*:*

$$r^* \le \sqrt{\frac{(K - K_c)(2\gamma_{\max} + K)^2}{K^3}}.$$

If additionally $\int (1/\gamma^3)\, \mathrm{d}\mu < \infty$*:*

$$\sqrt{\frac{8(K - K_c)}{K^3 K_c \int (1/\gamma^3)\, \mathrm{d}\mu}} \le r^*.$$

Proof sketch. At the fixed point $S(r^*) = 1$, i.e., $\int K/(\gamma + D)\, \mathrm{d}\mu = 1$, while $S(0^+) = K/K_c$.

Upper bound. The difference $S(0^+) - S(r^*) = K/K_c - 1 = (K - K_c)/K_c$. Since $D \le \gamma + Kr^*$, we have $K/(2\gamma) - K/(\gamma + D) \ge K^2 r^*/(2\gamma(\gamma + Kr^*))$. Under (H4), bounding below by $K^2 r^*/(2\gamma_{\max}(2\gamma_{\max} + K))$ and integrating yields $(K - K_c)/K_c \ge K^2 r^*/(K_c \gamma_{\max}(2\gamma_{\max} + K))$. Rearranging and using $r^{*2} \le r^*$ (since $r^* < 1$) gives the stated bound. The full calculation is carried out in `CriticalExponent.lean` (declaration `rstar_le_sqrt_gap`).

Lower bound. A Taylor expansion of $1/(\gamma + D)$ around $r = 0$ gives $K/(\gamma + D) = K/(2\gamma) - K^3 r^2/(4\gamma^3) + O(r^4)$. The leading correction term, bounded using $\int \gamma^{-3}\, \mathrm{d}\mu$, yields the lower bound. See `rstar_ge_sqrt_gap` in `CriticalExponent.lean`. □

Theorem 5.2 (Critical slowing down)**.** *Under* (H4)*, for* $K > K_c$ *with fixed point* $r^* \in (0, 1)$*:*

$$1 - \lambda \ \le\ \frac{(K - K_c)(2\gamma_{\max} + K)^2}{4\gamma_{\min}^3}. \tag{15}$$

In particular, $1 - \lambda = O(K - K_c)$ *as* $K \to K_c^+$*.*

Proof. From the upper bound in (14) and $D_{\min} \ge \gamma_{\min}$:

$$1 - \lambda \le \frac{K^3 r^{*2}}{D_{\min}(\gamma_{\min} + D_{\min})^2} \le \frac{K^3 r^{*2}}{\gamma_{\min}(2\gamma_{\min})^2} = \frac{K^3 r^{*2}}{4\gamma_{\min}^3}.$$

From Theorem 5.1: $r^{*2} \le (K - K_c)(2\gamma_{\max} + K)^2/K^3$. Substituting:

$$1 - \lambda \le \frac{K^3}{4\gamma_{\min}^3} \cdot \frac{(K - K_c)(2\gamma_{\max} + K)^2}{K^3} = \frac{(K - K_c)(2\gamma_{\max} + K)^2}{4\gamma_{\min}^3}.$$ □

Corollary 5.3. *The number of iterates to reach accuracy ε from above is at least $\lceil |\log \varepsilon|/(1-\lambda)\rceil$, which by* (15) *diverges at least on the order of $(K-K_c)^{-1}$ as $K \to K_c^+$ under the bounded-γ assumption* (H4)*.*

Theorem 5.4 (Critical slowing down — lower bound)**.** *Under* (H4) *and additionally $\int(1/\gamma^3)\,\mathrm{d}\mu < \infty$, for $K > K_c$ with fixed point $r^* \in (0,1)$:*

$$\frac{8(K-K_c)}{(\gamma_{\max}+K)(2\gamma_{\max}+K)^2 K_c \int(1/\gamma^3)\,\mathrm{d}\mu} \;\leq\; 1-\lambda. \tag{16}$$

Combined with (15)*: $1-\lambda = \Theta(K-K_c)$ as $K \to K_c^+$.*

Proof. From the lower bound in Theorem 4.4: $1-\lambda \geq K^3 r^{*2}/\big((\gamma_{\max}+K)(2\gamma_{\max}+K)^2\big)$ (since $D_{\max} \leq \gamma_{\max}+K$ and $\gamma_{\max}+D_{\max} \leq 2\gamma_{\max}+K$). From the lower bound in Theorem 5.1: $r^{*2} \geq 8(K-K_c)/(K^3 K_c \int \gamma^{-3}\,\mathrm{d}\mu)$. Substituting and canceling K^3 gives the result. ☐

6 Geometric convergence of Picard iterates

Define the Picard iteration: given $r_0 > 0$, set $r_{n+1} = \Phi(r_n)$.

6.1 Convergence from above

Theorem 6.1 (One-step contraction above r^*)**.** *For $r > r^* > 0$ with $\Phi(r^*) = r^*$:*

$$\Phi(r) - r^* \leq \lambda \cdot (r - r^*).$$

Proof. Apply Theorem 3.7 pointwise with $r_1 = r^*$ and $r_2 = r > r^*$:

$$\alpha^*(\gamma, K, r) - \alpha^*(\gamma, K, r^*) \leq \frac{K\gamma}{D^*(\gamma + D^*)} \cdot (r - r^*),$$

where $D^* = \sqrt{\gamma^2 + K^2 r^{*2}}$. The derivative is evaluated at r^* (the left endpoint), which bounds the difference quotient from above since $\partial_r \alpha^*$ is decreasing (Theorem 3.6). Integrating over μ:

$$\Phi(r) - \Phi(r^*) \leq \lambda \cdot (r - r^*).$$ ☐

Theorem 6.2 (Geometric convergence from above)**.** *Let $r^* > 0$ with $\Phi(r^*) = r^*$. For $r_0 > r^*$:*

$$r^* \leq r_n \leq r^* + \lambda^n (r_0 - r^*), \qquad \forall n \geq 0. \tag{17}$$

In particular, $r_n \to r^*$ as $n \to \infty$.

Proof. The lower bound $r_n \geq r^*$ holds by induction: Φ is strictly increasing (Proposition 2.6(b)), so $r_n \geq r^*$ implies $r_{n+1} = \Phi(r_n) \geq \Phi(r^*) = r^*$. The upper bound $r_n - r^* \leq \lambda^n (r_0 - r^*)$ follows by induction from Theorem 6.1. Since $0 < \lambda < 1$ (Theorem 4.2), $\lambda^n \to 0$. ☐

6.2 Convergence from below

Theorem 6.3 (Monotone convergence from below)**.** *For $0 < r_0 < r^*$ with $\Phi(r^*) = r^*$:*
- *(a) The iterates are monotone increasing: $r_n \leq r_{n+1}$ for all n.*
- *(b) The iterates are bounded above: $r_n \leq r^*$ for all n.*
- *(c) $r_n \to r^*$.*

Proof. Parts (a) and (b) follow from the sign lemma and monotonicity of Φ.

Part (a): By Lemma 3.3(10), $r_0 < r^*$ implies $\Phi(r_0) > r_0$, i.e., $r_1 > r_0$. If $r_n \le r_{n+1}$, then $r_{n+1} = \Phi(r_n) \le \Phi(r_{n+1}) = r_{n+2}$ by monotonicity of Φ, completing the induction.

Part (b): $r_0 \le r^*$ and Φ is monotone increasing imply $r_1 = \Phi(r_0) \le \Phi(r^*) = r^*$. By induction: $r_n \le r^*$ implies $r_{n+1} \le r^*$.

Part (c): By (a) and (b), (r_n) is increasing and bounded, hence converges to a limit $L \in [r_0, r^*]$. Since Φ is continuous on $(0,\infty)$ (being an integral of continuous functions bounded in $[0,1]$), $L = \Phi(L)$. If $L < r^*$, then $\Phi(L) > L$ by Lemma 3.3(10), contradicting $L = \Phi(L)$. Hence $L = r^*$. □

Theorem 6.4 (Geometric convergence from below). *For $0 < r_0 < r^*$ with $\Phi(r^*) = r^*$, define $\lambda_0 = \Phi'(r_0) = \int K\gamma/(D_0(\gamma + D_0))\,\mathrm{d}\mu$ where $D_0 = \sqrt{\gamma^2 + K^2 r_0^2}$. If $\lambda_0 < 1$, then:*

$$r^* - \Phi^n(r_0) \;\le\; \lambda_0^n\,(r^* - r_0) \quad \textit{for all } n \ge 0. \tag{18}$$

Proof. Since $\partial_r \alpha^*(\gamma, K, r)$ is decreasing in r (the tight Lipschitz property), and the iterates $r_n = \Phi^n(r_0)$ satisfy $r_0 \le r_n < r^*$ (Theorem 6.3), we have $\Phi'(r_n) \le \Phi'(r_0) = \lambda_0$ for all n.

The one-step bound: for $r < r^*$, the mean value theorem applied to $\alpha^*(\gamma, K, \cdot)$ on $[r, r^*]$ gives

$$\alpha^*(\gamma, K, r^*) - \alpha^*(\gamma, K, r) \;\le\; \frac{K\gamma}{D_r(\gamma + D_r)}\,(r^* - r)$$

where $D_r = \sqrt{\gamma^2 + K^2 r^2}$, using that $\partial_r \alpha^*$ is decreasing and hence bounded above by its value at the *left* endpoint. Integrating over μ:

$$r^* - \Phi(r) \;\le\; \Phi'(r)\,(r^* - r).$$

By induction: $r^* - r_{n+1} \le \Phi'(r_n)(r^* - r_n) \le \lambda_0(r^* - r_n) \le \lambda_0^{n+1}(r^* - r_0)$. □

Remark 6.5 (Comparison of rates from above and below). *From above, the rate is $\lambda = \Phi'(r^*)$. From below, it is $\lambda_0 = \Phi'(r_0) > \lambda$ (since Φ' is decreasing and $r_0 < r^*$). This reflects the genuine asymmetry: the derivative is larger on (r_0, r^*) than at r^*. As $r_0 \to r^{*-}$, the rate $\lambda_0 \to \lambda$ and the two bounds coincide.*

7 Subcritical regime

7.1 The linear bound

Theorem 7.1 (Linear bound on Φ). *Under (H1)–(H3), for all $r > 0$:*

$$\Phi(r) \le \frac{K}{K_c} \cdot r. \tag{19}$$

Proof. From the rationalized form and $D \ge \gamma$:

$$\alpha^*(\gamma, K, r) = \frac{Kr}{\gamma + D} \le \frac{Kr}{2\gamma}.$$

Integrating: $\Phi(r) \le (Kr/2)\int(1/\gamma)\,\mathrm{d}\mu = (K/K_c)r$. □

7.2 Strictly subcritical decay

Theorem 7.2 (Subcritical geometric decay). *For $K < K_c$ and any $r_0 > 0$:*

$$0 < \Phi^n(r_0) \le \left(\frac{K}{K_c}\right)^n r_0 \;\xrightarrow{n\to\infty}\; 0. \tag{20}$$

Proof. By induction from Theorem 7.1: $\Phi^{n+1}(r_0) \le (K/K_c)\cdot\Phi^n(r_0)$. Since $K/K_c < 1$, the bound converges to zero. Positivity holds because Φ maps $(0,\infty)$ into $(0,1)$ (Proposition 2.6(a)). □

7.3 The critical case $K = K_c$

Proposition 7.3 (Convergence at criticality)**.** *For $K = K_c$: $\Phi(r) < r$ for all $r > 0$, and $\Phi^n(r_0) \to 0$ for any $r_0 > 0$, but no exponential rate holds.*

Proof. The strict inequality $\Phi(r) < r$ follows from Theorem 3.5(a). The sequence $(\Phi^n(r_0))$ is strictly decreasing and bounded below by 0, hence converges to a limit $L \geq 0$. If $L > 0$, continuity of Φ on $(0, \infty)$ gives $L = \Phi(L) < L$, a contradiction. Hence $L = 0$. □

8 Contraction rate bounds

Theorem 8.1 (Upper bound on contraction rate)**.** *Under (H1)–(H3) and the additional assumption*

$$\int \gamma \, \mathrm{d}\mu < \infty \quad \textit{(finite mean)}, \tag{H5}$$

for any $r > 0$:

$$\lambda(r) = \int \frac{K\gamma}{D(\gamma + D)} \, \mathrm{d}\mu \leq \frac{\mathbb{E}_\mu[\gamma]}{Kr^2}.$$

Proof. Pointwise: $K\gamma/(D(\gamma + D)) \leq K\gamma/D^2 = K\gamma/(\gamma^2 + K^2r^2) \leq \gamma/(Kr^2)$, the last step since $K^2r^2 \leq \gamma^2 + K^2r^2$. Integrating gives the result. □

Remark 8.2. *At the fixed point: $\lambda(r^*) \leq \mathbb{E}[\gamma]/(Kr^{*2})$. Deep in the supercritical regime ($K \gg K_c$, $r^* \to 1$), this gives $\lambda \leq \mathbb{E}[\gamma]/K$: convergence accelerates with coupling strength.*

9 Distribution sensitivity

Theorem 9.1 (Monotonicity of K_c)**.** *If $\gamma_1(\omega) \leq \gamma_2(\omega)$ for all ω, then $K_c(\gamma_1) \leq K_c(\gamma_2)$.*

Proof. $1/\gamma_1 \geq 1/\gamma_2$ pointwise implies $\int (1/\gamma_1) \, \mathrm{d}\mu \geq \int (1/\gamma_2) \, \mathrm{d}\mu$, hence $K_c(\gamma_1) \leq K_c(\gamma_2)$. □

Theorem 9.2 (Monotonicity of r^*)**.** *If $\gamma_1 \leq \gamma_2$ pointwise and both have supercritical fixed points r_1^*, r_2^* at coupling K, then $r_2^* \leq r_1^*$.*

Proof. Suppose for contradiction that $r_1^* < r_2^*$. Since α^* is decreasing in γ (Proposition 2.6(b)), $\Phi_{\gamma_1}(r_2^*) \geq \Phi_{\gamma_2}(r_2^*) = r_2^*$. But $r_2^* > r_1^*$ and Lemma 3.3(11) (applied to Φ_{γ_1} with fixed point r_1^*) gives $\Phi_{\gamma_1}(r_2^*) < r_2^*$, a contradiction. □

Theorem 9.3 (Scaling)**.** *$K_c(c\gamma) = c \cdot K_c(\gamma)$ for $c > 0$.*

10 Lorentzian verification

For the Cauchy–Lorentz distribution with width $\gamma_0 > 0$, the OA residue evaluation gives constant dissipation $\gamma = \gamma_0$. Then $K_c = 2\gamma_0$ and $r^* = \sqrt{(K - 2\gamma_0)/K}$ for $K > 2\gamma_0$.

Proposition 10.1 (Lorentzian contraction rate)**.** *For the Lorentzian with $K > 2\gamma_0$:*

$$D = \sqrt{\gamma_0^2 + K^2 r^{*2}} = \sqrt{(K - \gamma_0)^2} = K - \gamma_0, \tag{21}$$

$$\lambda = \frac{K\gamma_0}{D(\gamma_0 + D)} = \frac{K\gamma_0}{(K - \gamma_0) \cdot K} = \frac{\gamma_0}{K - \gamma_0}, \tag{22}$$

$$1 - \lambda = \frac{K - 2\gamma_0}{K - \gamma_0}. \tag{23}$$

Proof. We compute $D^2 = \gamma_0^2 + K^2(K - 2\gamma_0)/K = \gamma_0^2 + K^2 - 2K\gamma_0 = (K - \gamma_0)^2$. Since $K > 2\gamma_0 > \gamma_0$, $D = K - \gamma_0 > 0$. Then $\gamma_0 + D = K$, giving (22)–(23). $\square$

Remark 10.2 (Consistency checks)**.**

1. Spectral gap formula*:* $K^3 r^{*2}/(D(\gamma_0 + D)^2) = K^3 \cdot (K - 2\gamma_0)/(K \cdot (K - \gamma_0) \cdot K^2) = (K - 2\gamma_0)/(K - \gamma_0) = 1 - \lambda$. ✓
2. Critical slowing down*: as* $K \to 2\gamma_0^+$, $1-\lambda = (K-2\gamma_0)/(K-\gamma_0) \sim (K-K_c)/\gamma_0$*, confirming linear vanishing with coefficient* $1/\gamma_0 = 2/K_c$*.* ✓
3. Strong coupling*: as* $K \to \infty$, $\lambda = \gamma_0/(K - \gamma_0) \to 0$*, confirming fast convergence.* ✓

11 The main theorem

Theorem 11.1 (Quantitative phase transition)**.** *Let* μ *be a probability measure on* $(\Omega, \mathcal{F})$ *and* $\gamma : \Omega \to (0, \infty)$ *measurable with* $\int(1/\gamma)\,\mathrm{d}\mu \in (0, \infty)$*. Set* $K_c = 2/\int(1/\gamma)\,\mathrm{d}\mu$*.*

Subcritical *(*$K < K_c$*)**:*** $\Phi(r) < r$ *for all* $r > 0$*. No positive fixed point exists. Picard iterates decay exponentially:* $\Phi^n(r_0) \leq (K/K_c)^n r_0 \to 0$*.*

Critical *(*$K = K_c$*)**:*** $\Phi(r) < r$ *for all* $r > 0$*. No positive fixed point exists.* $\Phi^n(r_0) \to 0$ *by monotone convergence; no exponential rate is claimed.*

Supercritical *(*$K > K_c$*)**:*** *There exists a unique* $r^* \in (0, 1)$ *with* $\Phi(r^*) = r^*$ *(Theorem 3.5). The derivative at the fixed point is*

$$\lambda = \Phi'(r^*) = \int \frac{K\gamma}{D_\gamma(\gamma + D_\gamma)}\,\mathrm{d}\mu \in (0, 1),$$

with spectral gap

$$1 - \lambda = \int \frac{K^3 r^{*2}}{D_\gamma(\gamma + D_\gamma)^2}\,\mathrm{d}\mu > 0.$$

For $r_0 > r^*$*: geometric convergence* $r^* \leq r_n \leq r^* + \lambda^n(r_0 - r^*)$*.*
For $0 < r_0 < r^*$*: monotone convergence* $r_n \nearrow r^*$ *(no geometric rate claimed).*

12 Proof architecture and formal verification

12.1 Repository and build information

The Lean 4 formalization is available at:

`https://github.com/taejun-song/kuramoto-lean`
commit `e86b442` · Lean `v4.30.0-rc1` · Mathlib `v4.30.0-rc1`

To reproduce: clone the repository, run `lake build`. All declarations below compile with 0 `sorry` and no project-specific axioms (only the standard Lean 4/Mathlib foundations: propositional extensionality, quotients, and choice).

12.2 Lean file structure

File	Content	Status
`TightLipschitz.lean`	Spectral gap, geometric convergence	0 sorry
`SelfConsistencyContraction.lean`	Sign lemma, contraction, map range	0 sorry
`ContinuumBifurcationDiagram.lean`	Existence, uniqueness, bifurcation	0 sorry
`CriticalExponent.lean`	$r^* = \Theta(\sqrt{K - K_c})$ bounds	0 sorry
`IterationConvergence.lean`	Picard monotonicity and boundedness	0 sorry
`ContractionRate.lean`	Quantitative contraction	0 sorry
`QuantitativeRefinements.lean`	Two-sided CSD, geometric below	0 sorry
`DistributionSensitivity.lean`	K_c, r^* comparison across γ	0 sorry
`UnifiedPhaseTransition.lean`	Combined packaging theorem	0 sorry

12.3 Correspondence between paper and Lean declarations

Paper result	Lean declaration
Lemma 3.2 (slope identity)	`slope_integral_eq_one`
Lemma 3.3 ($\Phi(r) > r$ below)	`sc_map_exceeds_below`
Lemma 3.3 ($\Phi(r) < r$ above)	`sc_map_below_above`
Theorem 3.5 (bifurcation)	`kuramoto_phase_transition`
Theorem 3.5(b) (uniqueness)	`continuum_exactly_one_equilibrium`
Theorem 3.6 (tight derivative)	`explicitEquil_tight_lipschitz`
Theorem 4.2 ($\lambda < 1$)	`tight_lipschitz_lt_one`
Theorem 4.3 (gap identity)	`spectral_gap_integral`
Theorem 4.4 (lower bound)	`spectral_gap_lower_bound`
Theorem 4.4 (upper bound)	`spectral_gap_upper_bound`
Theorem 5.2 (critical slowing)	`critical_slowing_down`
Theorem 5.4 (CSD lower)	`critical_slowing_down_lower`
Theorem 6.1 (one-step)	`sc_map_tight_contraction_above`
Theorem 6.2 (geometric)	`scMapIter_geometric_above`
Theorem 6.2 (topological)	`scMapIter_tendsto_above`
Theorem 6.3 (from below)	`scMapIter_converges_below`
Theorem 6.4 (geometric below)	`scMapIter_geometric_below`
Theorem 7.1 (linear bound)	`sc_map_linear_bound`
Theorem 7.2 (subcritical)	`subcritical_geometric_decay`
Theorem 8.1 (rate bound)	`contraction_rate_upper`
Theorem 9.1 (K_c monotone)	`continuumKc_mono_gamma`
Theorem 9.2 (r^* monotone)	`rstar_mono_gamma`

12.4 Key formalization decisions

1. **Abstract probability space.** Theorems are stated over $(\Omega, \mathcal{F}, \mu)$ with $\gamma : \Omega \to \mathbb{R}$ satisfying $\gamma(\omega) > 0$ pointwise.
2. **Measurability via level sets.** The Lean formalization assumes $\forall M$, $\{\omega \mid \gamma(\omega) \leq M\}$ is measurable (implying Borel measurability of γ).
3. **Integrability by dominance.** The integrability of $K\gamma/(D(\gamma + D))$ is proved by showing it is pointwise $< K/(\gamma + D)$ (Theorem 3.8), and the latter is integrable by the fixed-point identity.
4. **Spectral gap by subtraction.** The formula (13) is derived by integrating the *difference* of the slope $K/(\gamma+D)$ and the derivative $K\gamma/(D(\gamma+D))$, using the linearity of integration.

5. **Topological convergence.** The Lean proof of $r_n \to r^*$ from above uses Mathlib's `Filter.Tendsto` framework and the squeeze theorem, not just the geometric bound.

13 Discussion

13.1 Summary

The central contribution is the spectral gap identity (13): $1 - \Phi'(r^*) = \int K^3 r^{*2}/(D(\gamma+D)^2)\,\mathrm{d}\mu$. This exact formula reveals how the gap depends on the coupling K (numerator $\sim K^3$) and the frequency dispersion (denominator controlled by γ through D). The two-sided critical slowing down bounds (15)–(16) establish $1 - \lambda = \Theta(K - K_c)$, making rigorous the physics intuition that convergence becomes arbitrarily slow near onset. The geometric bound from below (Theorem 6.4) shows that convergence to r^* is exponential from *any* starting point, not only from above.

13.2 Relation to PDE stability

The self-consistency map Φ governs the *static* fixed-point equation. The *dynamical* convergence $r(t) \to r^*$ for the ODE/PDE system is studied by [12, 1, 4]. The two are related: the derivative $\Phi'(r^*)$ controls the linearized dynamics of $\dot{r} = \Phi(r) - r$ near r^*, giving exponential rate $1 - \lambda$. Whether $1 - \Phi'(r^*)$ equals the leading eigenvalue of the linearized PDE operator (as studied by Chiba) is an open question.

13.3 Computational applications

- **A priori error bound**: after n iterations from $r_0 > r^*$, $|r_n - r^*| \leq \lambda^n (r_0 - r^*)$.
- **Iteration count**: $n \geq \lceil \log((r_0 - r^*)/\varepsilon)/(-\log \lambda) \rceil$ iterations suffice for accuracy ε (for $r_0 > r^*$).
- **Subcritical detection**: $K_c = 2/\int \gamma^{-1} \mathrm{d}\mu$ is computable from the distribution. If $K/K_c < 1$, the linear bound (19) guarantees $\Phi(r) \leq (K/K_c)\, r$ for all $r > 0$, giving subcritical exponential decay with rate K/K_c.

13.4 Limitations

1. The geometric rate from above uses the fixed-point derivative $\lambda = \Phi'(r^*)$; from below, the rate $\lambda_0 = \Phi'(r_0) > \lambda$ depends on the initial point.
2. Critical slowing down requires bounded γ. For unbounded distributions, the identity (13) still holds but (15) does not.
3. We study the self-consistency map, not the full PDE. Connecting $\Phi'(r^*)$ to PDE eigenvalues requires additional analysis.

13.5 Open questions

1. Is the rate $\lambda_0 = \Phi'(r_0)$ from below optimal, or can it be improved (e.g., to $\Phi'(r^*)$ under additional regularity)?
2. What is the relationship between $1 - \Phi'(r^*)$ and Chiba's generalized eigenvalues [1]?
3. At $K = K_c$: the self-consistency equation $\Phi(r) \approx r - cr^3$ near $r = 0$ suggests $\Phi^n(r_0) = O(1/\sqrt{n})$. Can this be proved?

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